

Powers of Ten and Place Value

Answer Key

Grades 4–5 Mathematics

Quick Answers

1. $7 \times 100 = 700$, $7 \times 1,000 = 7000$
2. 260
3. 40000
4. 1000
5. 350
6. 620
7. 4500

8. 8
 9. how many times = 10, —
 10. value of the 5 in 52,600 = 50000, value of the 5 in 5,260 = 5000, how many times = 10
 11. 1000
 12. 6000
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What is verified

11 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 9. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics

4.NBT.A.1 – *problems 9, 10*

5.NBT.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 11, 12*

Difficulty 1–4 of 5 across 16 tagged checks.

- 1.** Complete the pattern: $7 \times 10 = 70$; 7×100 ; $7 \times 1,000$.

Solution: Multiplying by 10 shifts every digit one place left, so each line gains one more zero than the line before. $7 \times 100 = 700$

$$7 \times 1,000 = 7,000$$

700

7000

- 2.** 26×10

Solution: Shift each digit one place left: 2 tens become 2 hundreds and 6 ones become 6 tens. $26 \times 10 = 260$.

260

- 3.** $40 \times 1,000$

Solution: Multiply $4 \times 1,000 = 4,000$, then keep the factor of ten from 40: shift once more. $40 \times 1,000 = 40,000$.

40000

- 4.** $10^3 = 10 \times 10 \times 10$

Solution: The exponent 3 counts the factors of ten: three shifts of the digit 1, which writes a 1 followed by three zeros. 1000

5. $3,500 \div 10$

Solution: Dividing by 10 shifts every digit one place right: 3 thousands become 3 hundreds, 5 hundreds become 5 tens. 350

6. $62,000 \div 100$

Solution: Dividing by 100 shifts two places right — remove two zeros: $62,000 \div 100 = 620$. 620

7. 45×10^2

Solution: $10^2 = 100$, so shift the digits of 45 two places left: $45 \times 100 = 4,500$. 4500

8. $8,000 \div 10^3$

Solution: $10^3 = 1,000$, so shift three places right: $8,000 \div 1,000 = 8$. 8

9. In 66,000, the ten-thousands 6 is how many times the thousands 6? Explain.

Solution: The first 6 is worth 60,000; the second is worth 6,000. $60,000 \div 6,000 = 10$. 10

Full credit for the explanation: each place is worth *10 times* the place to its right, and the ten-thousands place sits one place to the left of the thousands place — so the same digit is worth 10 times as much there.

10. The two 5s in 52,600 and 5,260.

Solution: In 52,600 the 5 is in the ten-thousands place: $5 \times 10,000 = 50,000$. 50000

In 5,260 the 5 is in the thousands place: $5 \times 1,000 = 5,000$. 5000

The digit moved one place, so the first value is $50,000 \div 5,000 = 10$ times the second. 10

11. How many times as large as 10^2 is 10^5 ?

Solution: $10^5 = 100,000$ and $10^2 = 100$. Dividing, $10^5 \div 10^2 = 100,000 \div 100 = 1,000$ — three factors of ten remain, which matches subtracting the exponents: $5 - 2 = 3$. 1000

12. One box holds 10 markers; one case holds 100 boxes. Markers in 6 cases?

Solution: One case holds $100 \times 10 = 1,000$ markers (a box of boxes multiplies the powers of ten). Six cases hold $6 \times 1,000 = 6,000$ markers. 6000